



INTERCEPTS®

August

NEWSLETTER

1993

■■■ "CODE WORDS" ■■■

Military Codewords and Acronyms

MILSPEAK REVISITED SUMMER 93

Are you confused when your military buddies tell you they saw a warthog fly over your house? When you tune into the military frequencies do you only understand every other word and can't tell the difference between a BUFF and a SLUFF? Then welcome to the club!

Military communications monitoring is a tricky technical challenge calling for electronic skills, research and hours of patience. But even after mastering the military airwaves it can be hard to understand what is going on. Military communications are not only cryptic but technical as well and it's hard to follow the action when you can't understand the military jargon.

So here we go again. It's time for our annual military jargon (Milspeak) update. This year we have included some acronyms that you might hear while monitoring the military. You can't follow the game without a program so lets get to it.

Code Word/Acronym	Definitions
AACOMS	Army Area Communications System
ABDR	Aircraft Battle Damage Repair
ABM	Anti-Ballistic Missile
ACE	Air Combat Element
ACE HIGH	NATO tropospheric scatter
ACLS	Automatic Carrier Landing System
ACM	Air Combat Maneuvers
ACSI	Assistant Chief Of Staff, Intelligence
ACT	Air Combat Tactics
ADIZ	Air Defense Identification Zone
AEGIS	Advanced Naval Air Defense Radars System
AFSATCOM	Air Force Satellite Communications
AGL	Above Ground Level
AICC	Airborne Intercept Command Communications
AIM-7	Sparrow radar-guided medium-range air-to air missile
AIM-9	Sidewinder infrared guided short-range air-to air missile
AIM-54	Phoenix radar-guided long-range air-to-air missile
ALARM	Air Launched Alert Response Missile; Aircraft Launched Space Booster
ALCC	Airborne Launched Control Center
ALCM	Air Launched Cruise Missile
ALFA HOTEL	Disagreeable pilot (asshole)
ALLIGATOR	AWACS secure voice
AMBER DECK	Flight deck standby condition
AMMO MINUS	Less than half ammunition left
AMMO PLUS	More than half ammunition
AMRAAM	AIM-120 Advanced Medium Range Air-to-Air Missile
ANCHORED	Navy aircraft orbiting area
ANGLES	Altitude-thousands of feet
ANVIS	Airmen's Night Vision Imaging System
AO	Aviation Officer
ARBS	Angle Rate Bombing System
AREA 51	Top secret military weapons test center located on the Nellis Air Force range
ASAT	Anti-Satellite weapon
ASIS	Attack Systems Interrogation Station (mission commander's station in B-2)
ASMP	Stealth Bomber, responsible for navigation and attack)
ASPJ	Mirage Launched Cruise Missile
ASR	Airborne Self Protection Jammer
ASROC	Approach Surveillance Radar
ASTOR	Anti-Submarine Rocket
ASW	Anti-submarine torpedo
ATB	Anti-Submarine Warfare
	Advanced Technology Bomber (B-2)
ATF	
ATIS	
ATO	
AUTODIN	
AUTOSEVCON	
AUTOVON	
AWACS	
AWLS	
BAG	
BALL	
BANDIT	
BAT TURN	
BDA	
BEACH	
BEACON DOUBLE	
BEAD WINDOW	
BEAM RIDER	
BENT	
BENT SPEAR	
BFM	
BINGO	
BINGO FIELD	
BIRD DOG	
BIRDS	
BLUE DECK	
BLUE FORCES	
BMB	
BMEW	
BN	
BOGEY	
BOLTER	
BRAVO ZULU	
BREAK	
BRICKWORK	
BROKEN ARROW	
BROTHER	
BROWNIE	
BUFF	
BURNER	
BUSTER	
C-3	
C3CM	
C31	
CAG	
CAP	
CAPC	
CARQUAL	
CASKET	
CATCC	
CAVU	
CAX	
C BAND	
CCD	
CERCOM	
CHARLIE	
CHATTER	
CHECK	
CHECK SIX	
CHERUBS	
CHEVRON	
CHICKS	
CHICKS IN TOW	
CHOP	
CIA	
CIC	
CINC	
Advanced Tactical Fighter	
Automatic Terminal Information System	
Airborne Tactical Officer	
Automatic Digital Network	
Automatic Secure Voice Communications	
Automatic Voice Network	
Aircraft Approach and Warning Control System	
All Weather Landing System	
Flight suit	
Optical landing device on carrier	
Hostile aircraft	
High-G tight turn	
Bomb Damage Assessment	
Desert-ground	
On your tracking beacon select double plus code	
Communications have been compromised	
Missile guided by radio or radar	
Inoperative	
Nuclear accident	
Basic Flight Maneuvers	
Minimum fuel for safe return	
For landing alternate airfield	
Surveillance aircraft	
Guided missiles or satellites	
Flight deck handling medical emergency	
Forces are friendly	
Ballistic Missile Defense	
Ballistic Missile Early Warning	
Bombardier-Navigator	
Enemy aircraft	
Unsuccessful carrier landing	
Job well done	
Sharp turn or interrupting communications	
A line of sonobuoys to detect submarine passage	
Nuclear weapon damage	
Attacking ship of surface ASW unit	
Photo mission	
B-52 bomber	
Afterburner	
Continuous maximum flying speed	
Command communications and control	
Command control communications and counter measure	
Command control communications and intelligence	
Commander of Air Group	
Combat Air Patrol	
Carrier Aircraft Plane Commander	
Carrier Qualifications (Also CQ)	
Aircraft crashing or out of control	
Carrier Air Traffic Control Center	
Clear Air Visibility Unlimited	
Combined Arms Exercise	
The radio frequency band of 3.9 to 6.2 gigahertz—band	
Charged Coupled Device	
Acronym for Army Communications and Electronics Material Readiness Command	
Correct	
Jamming communications	
Alter heading momentarily port/starboard	
Look for bogey behind plane	
Altitude in hundreds of feet	
Sonobuoy pattern	
Friendly fighter aircraft	
Friendly aircraft refueling	
Changing operational control to another activity	
Central Intelligence Agency	
Combat Information Center	
Acronym for Command in Chief, CINC	

CINCSpaceCOM	denotes the highest-ranking military officer in each of the United States military commands	FEET DRY	Aircraft over beach
CIRVIS	Commander In Chief, Space Command	FEET WET	Aircraft over water
CLARA	Comms/Instructions for reporting Visual Intelligence Sightings (UFOs)	FERRET	COMINT gathering satellite or aircraft
CLEAN	Radar scope clear	FEW	Seven or fewer aircraft
CODE 1	In-flight configuration without gear/flaps/ordnance extended	FIGHT'S ON	Air combat beginning
CODE 2	Aircraft in perfect working order	FLASH	Coded message-high priority
CODE 3	Aircraft with minor mechanical problems	FLIR	Forward Looking Infra-Red display
COLD	Aircraft with major mechanical problems	FLTSATCOM	Fleet Satellite Communications (system)
COLD NOSE	Not using live ordnance or lost ASW contact	FOBS	Fractional Orbiting Bombardment Systems
COLLATERAL DAMAGE	Aircraft operating with radar	FOD	Foreign Object Demand
COLOSSUS	Civilian casualties or untargeted destruction	FORSCOM	Forces Command
	The Sound Surveillance System (SOSUS) that is developed along the Pacific coast of the United States for the purpose of detecting Soviet submarines	FOULED DECK	Deck not ready to receive aircraft
COMINT	Communications Intelligence	FOX	Missile launching
COMIREX	Committee On Imagery	FOX AWAY	Missile fired or released from aircraft
COMOR	Requirements And Exploitation	FOX ONE	Launching sparrow missile
COMPAC	Committee On Overhead Reconnaissance	FOX TWO	Launching sidewinder missile
	Designation for man-portable SATCOM terminal Radar chaff	FOX THREE	Launching phoenix missile
COMSEC	Abbreviation for Communications Security	FRAG	On time
CONFETTI	Radar chaff	FRATRACIDE	Accidentally shooting at one's own troops
CORANDCOM	Abbreviation for Army Communications Research and Development Command	FREAK	Frequency in Mega Hertz
	Code word for nuclear secrets that are more restricted than top secret information	FREDDIE	Controlling unit
COSMIC	Hostile act confirmed	FREELANCE	Aircraft being self-controlled
COVERED WAGON	Pilot not ejecting in an emergency	FROG	Free (unguided) Rocket Over Ground
COWBOY	C.P. Command Post (military/police operation)	FSS	Flight Service Station
CRYSTAL BALL	Radar scope	FURBALL	Multi aircraft aerial engagement
CSOC	Consolidated Space Operation Center	GADGET	Radar equipment
CSS-4	Chinese nuke	GAS PASSER	Aerial refueling boom operator
CZ	Combat zone	GATE	For limited time fly at maximum speed
DARPA	Defense Advanced Research Projects Agency	GC	Green Coordination secure voice radio
DASC	Direct Air Support Center	GCHQ	Government Communications Headquarters (Britain)
DASH TWO	Second ship in two ship formation	GCI	Ground Control Intercept
DCI	Director of Central Intelligence	GEODSS	Ground-Based Electro-Optical Deep Space Surveillance System
DDP	Deputy Director of Plans (CIA)	GERTRUDE	Underwater sonar telephone
DDS&T	Deputy Director of Science And Technology (CIA)	GIB	Guy In Back (Electronics Systems Operator)
DEFCON	Defense Condition (ranges from 5 for peace to 1 all out war)	GOBLIN	Electronic Warfare Problem
DEFSMAC	Defense Special Missile And Astronautics Center	GOLDEN BB	Anti aircraft artillery
DELTA1/2/3/4	Russian Navy SSBN	GOLDIE	Flight control/bombing systems ready for ground control
DEP CON	Departure Control	GOMER	Slang for enemy
DEVILS	Depth	GONE ELVIS	Missing in action
DEW	Distant Early Warning	GPS	Global Positioning System
DF	Direction Finding equipment	GRAND SLAM	Destroyed all enemy aircraft
DIA	Defense Intelligence Agency	GREEN	Scrambled communications
DIP	Lowering sonar	GREEN DECK	Flight deck clear-launch and recover
DME	Distance Measuring Equipment	GUARD	International distress radio frequencies (121.500 MHz, 243.000 MHz, 406.000 MHz)
DOLLY	Airborne data link equipment	GULL	Deceptive floating radar reflector (Navy)
DOUBLE NUTS	Aircraft number ending in 00	HAC	Helicopter Aircraft Commander
DOUBLE UGLY	F-4 Phantom	HACK	Aircraft used-ferrying pilots
DOWN LINK	The rebroadcast signal from a satellite	HANGER QUEEN	Maintenance problems-aircraft used for parts
DRONE	Remotely piloted aircraft	HAPPY HOUR	Return to base-activity complete
DRR	Departure Reference Radial	HARM	High-speed Anti-Radiation Missile
DSCS	Defense Satellite Communications Systems	HAVE QUICK	Frequency hopping UHF radio system
DSMP	Defense Satellites Meteorological Program	HAWKEYE	Gruman E2-C aircraft
DSP	Defense Support Program (early warning satellite)	HEADING	Bogey's magnetic course is
DUCK	Headed your way	HEADS UP	Enemy approaching
DULL SWORD	Nuclear incident	HELPING HAND	Threat of hostile act
DZ	Drop Zone	HF	High frequency 1.6 to 30 MHz
EAM	Emergency Action Message	HIGH	Altitude between 25-50,000 ft.
ECM	Electronic Counter Measures	HOLDING HANDS	Flying in-formation
EEAT	Emergency Expected Approach Time	HOMEPLATE	Aircraft's home base
EHF	Extremely High Frequency	HOT	Using live ordnance or active ASW contact
ELECTRIC JET	F-16 Falcon	HOTAS	Hands On Throttle And Stick
ELEVATOR	Altitude indicated in 5000 intervals	HTB	High Threat Battlefield
ELF	Extremely Low Frequency 30 to 300 Hz	HUD	Heads Up Display
ELINT	Electronic Intelligence	HUMMER	Personnel carrier
ELT	Emergency Locator Transmitter	HUMINT	Human Intelligence gathering (spying)
EMCON	Emergency Condition Emission Control	HUNKER	Prepare for defensive action and maximize terrain masking
EMP	Electromagnetic Pulse burst of energy (following nuclear explosion)	ICAO	International Carrier Air Operators
EOD	Explosive Ordinance Disposal (unit)	ICBM	Inter-Continental Ballistic Missile
ERCS	Emergency Rocket Communication System	IDENTIFY	Identify the contact
EXCOM	Executive Committee	IFE	In Flight Emergency
FAA	Federal Aviation Administration	IFR	Instrument Flight Regulations
FAC	Forward Air Controller	IMC	Instrument Maneuvering Condition
FACSFAC	Fleet Air Control And Surveillance Facilities	INDICATING	Contact speed by plot is
FADED	Contact disappeared from scope	INR	Bureau Of Intelligence And Research
FADED GIANT	Radiological incident	INS	Inertial Navigation System
FAMISHED	Are there any instructions please?	IN THE DARK	Not visible on radar display
FAST MOVERS	Jet aircraft	IP	Initial Point or Intercept Point
FATHER	Activated TACAN	IR	In Flight Route Or Infrared
FEBA	Forward Edge of Battle Area	I STAY	Remaining on patrol/mission_hours
CODEWORDS		JAMMING	Green:clear amber:partial red:complete blue:total
		JANAP	Joint Army Navy Aircraft Procedures
		JANET	DOE personnel ferrying aircraft
		JCS	Joint Chiefs Of Staff
		JOKER	Air contact not hostile
		JPL	Jet Propulsion Laboratories
		JRC	Joint Reconnaissance Center
		JSTARS	Joint Surveillance Targeting Acquisition Radar System
		JUDY	Have contact and taking over intercept

JUMP	Combat disengage	OTC	Officer and Tactical Command
KH	Keyhole intelligence satellite	PACCS	Post Attack Command and Control
KILO	Frequency change		System
KNOCK IT OFF	Maneuvers or combat-disengage	PACER SKY	Satellite based reconnaissance
LAMPS	Light Airborne Multi-Purpose Systems		system that transmits real-time
LANTERN	Turn Low Altitude Navigation and		infrared, visual, and radar data
	Targeting Infra-Red for Night system	PACK	Passengers on board
LASA	Laser Anti-Satellite weapon	PAINT	Radar indication
LASERCOM	Laser Communications (systems)	PAN	Safety Message-very urgent
LAZY	Equipment on standby	PARROT	I.F.F. equipment
LASINT	Laser Intelligence (information	PAVE PAWS	Phased-array radars
	gathered on foreign laser capabilities	PET ROCKS	Nuclear weapons (unused)
L-BAND	Designation for 390 to 1550 MHz	PHOTINT	Photographic Intelligence
LEVEL	Upper altitude is at designated contact	PICTURE CLEAN	Radar display-no unfriendly aircraft
LF	Low Frequency	PIGEONS	Magnetic bearing and distance from
LGM-30F	Minuteman II Ballistic Missile		you is _____
LGM-30G	Minuteman III Ballistic Missile	POINT FAILURE	Malfunctioning satellite
LINER	Fly at speed-maximum cruising range	PINGER	Active sonobuoy
LOCALIZE KILO	Establish area-Kilo-class sub operators	PLAYGROUND	Exercise area
LOUD HANDLE	Releases ejection seat	PLAYMATE	Aircraft involved in same mission
LOW	Altitude between 500-2000 ft.	PLAYTIME	Commence aerial maneuvers
LSO	Landing Signal Officer (aircraft carriers)	PLRS	Position Location Reporting System
LZ	Landing Zone	PLSS	Position Location Strike System
MACH	Speed of sound	PLUG	Connection with aerial refueler
MAD	Magnetic Anomaly Detection	POB	Persons On Board
	(submarine hunters)	POGO	Return to frequency
MAG	Marine Aviation Group	POLARIS	US Navy Ballistic Missile
MANY	Eight or more aircraft-raid size	POPEYE	Reduced visually
MARSA	Military Assumes Responsibility For	POPPED	Killed destroyed
	Separation	POSEIDON	US Navy ballistic missile
MARSHALL	Gathering point for approach aircraft	POSER	Pilot all talk no action
MAW	Marine Aviation Wing	POUNCE	In position to engage target
MAYDAY	Distress call	PUNCH	Intercept target very soon
MAWTS	Marine Aviation Weapons And	PURPLE	Unit suspected in carrying nuclear
	Tactics Squadron		weapons
MCS	Maneuver Control System	PURPLE FORCES	Opposing forces-blue & orange
MEDIUM	Altitude between 20-5,000 ft.		(usually subs & aircraft)
MEGA FORTRESS	EB-52 Strategic "battleship" escort aircraft	PTP	Permission To Proceed
MERGE	When two aircraft or tracks come	Q-BAND	Designation for 36,000 to 46,000 MHz
	together	RACKET	Radar chaff
METRO	Weather	RED DECK	Flight deck not clear
MEUSOC	Marine Expeditionary Unit Special	RADINT	Radar intelligence
	Operation Capable	RAF	Royal Air Force
MF	Medium Frequency	RAIN	Radar chaff
MGM-118A	Peacekeeper Ballistic Missile	RC	Red Communications (no scrambling)
MICKEY	Minutes To Go	RCR	Runway Condition Report
MIDAS	Missile Defense Alarm System	REMF	Rear-Eschelon (Mother Fu—ers)
MIDDLEMAN	Radio relay equipment-VHF or UHF		troops in the rear areas
MIGCAP	Combat air patrol over ground	RESCAP	Rescue combat and patrol
	attack aircraft	RESUME	Resume last patrol ordered
MIKES	Minutes	RIO	Radar Intercept Officer
MIRV	Multiple independently targetable	ROAD	Line of sonobuoys
	Re-entry Vehicle (nukes)	ROGER	Message received and understood
MIXUP	A mix of friendly and hostile aircraft	RON	Remain overnight
MOA	Military Operations Area	RORSAT	Radar Ocean Reconnaissance Satellite
MOTHER	Pilot's destination-aircraft carrier	RSO	Reconnaissance Systems Officer
MRBM	Medium Range Ballistic Missile	RTB	Return To Base
MRE	Meals Ready to Eat, jokingly called	S-BAND	Radio frequency band of 1550 to 5200 MHz
	meals rejected by Ethiopians	SACA	Supporting Arms Coordinator to Artillery
MUSHROOM	Special weapon (nukes?)	SAM	Surface to Air-Missile
MUSIC	Electronic jamming	SAM (plus number)	Special Air Mission usually denotes
NALF	Naval Auxiliary Landing Field		dignitary, politician or high ranking
NARF	Naval Air Rework Facilities		military is onboard
NAS	Naval Air Station	SAMOS	Satellite And Missile Observation
NATO	North Atlantic Treaty Organization	SANDBAGS	Dug in troops
NATOPS	Naval Training Operation	SAND-BOY EXPRESS	Ground transport
NAVAID	Navigational Aid	SAUNTER	Fly at best fuel consumption
NCA	National Command Authorities	SCAN	Search sector indicated & report contacts
NDS	Nuclear Detection System	SCF	Satellite Control Facility
NEST	Nuclear Emergency Search Team	SCI	Sensitive Compartmented Information
NOIC	Navy Operational Intelligence Center	SCOOTER	A-4 Skyhawk
NO JOY	No contact	SCRAM	Keep clear, about to open fire
NOLF	Naval Outline Landing Field	SCRUB	Abort mission
NO LONE ZONE	Prohibited area	SCUD	Iraqi long range missile
NORAD	North American Aerospace Defense	SCUD-BUSTER	Patriot missile
	Command	SEAD	Suppression of Enemy Air Defenses
NREC	National Reconnaissance Executive	SEEK TALK	Designation for secure-
	Committee		communications system
NRF	Naval Radio Facility	SENSO	Sensor operator on S-3 Wing
NRL	Naval Research Laboratory	SHF	Superhigh frequency
NRO	National Reconnaissance Office	SHINE	Radar or sonar-target is clear
NSA	National Security Agency	SI	Special Intelligence
NSC	National Security Council	SICK	Equipment problems
NSP	Navy Space Project	SIERRA HOTEL	Polite phonetics for shit hot
NYLON LETDOWN	Return by parachute	SINKER	Radar contact disappearing
ONR	Office of Naval Research	SKIP IT	Cease activity
ON TARGET	Fire control director/system	SKUNK	Unidentified surface contact assumed
	acquired/tracking OK		hostile
O. P.	Orbiting Point	SIGINT	Signal Intelligence (COMINT, ELINT,
ORANGE FORCES	Exercise-enemy role forces		TELINT)
ORANGES SOUR	Weather unsuitable	SLAR	Side-Looking Airborne Radar
ORANGES SWEET	Weather suitable	SLBM	Submarine Launched Ballistic Missile
ORBITING	Circle and search or circling	SLOM	Submarine (Sea) Launched Cruise Missile
OSD	Office of Secretary of the Defense	SLEP	Service Life Extension Program
OSCAR	Optical Submarine Communications	SNOW	Sweep jamming
	by Aerospace Relay	SOB (s)	Souls On Board
OSO	Office of SIGINT Operations	SOP	Standard Operating Procedure

SORTIE SORTS	Aerial combat mission Status of operational readiness and training system
SOSUS	Sound surveillance system (ASW)
SOUR	Radio contact poor
SPADATS	Space tracking and detection system
SPADOC	Space Defense Operation Center
SPIE	Special insertion and extraction (seals)
SPIT CADILLACS	Drop sonobouys
SPLASH	Shot down aircraft into sea
SPOOFER	Contact employing electronic or tactical jamming
SPOOK	Possible enemy submarine
SQUAWK	Send identification friend or foe at code/mode indicated
SR	Strategic Reconnaissance
SR-75 aircraft	Hypersonic SR-71 possible replacement
SRAM	Short Range Attack Missile
SSBN	Ballistic Missile Submarine
SS-11	Russian nuke
SS-13	Russian nuke
SS-18	Russian nuke
SS-19	Russian nuke
SS-24	Russian nuke rail-mobile missile
SS-25	Russian nuke
SSBS S-3	French nuke
SSTS	Space Surveillance Tracking System
STARCOM	Strategic Army Communications
START CART	Engine starting compressor
STATE CHICKEN	Fuel state requires recover, refueling, or diversion
STATE LAMB	Inadequate fuel for mission & reserve to return
STATE TIGER	Sufficient fuel for mission & reserve to return
STEADY	Straighten cut heading or on prescribed heading
STEER	Indicate to set magnetic heading
STRANGER	Unidentified aircraft
STRANGLE	Radio silence
SWEET	Good contact
SWEET LOCK	Good contact & locked on signal
TACAMO	Designation for airborne strategic Communications System also known as the airborne radio relay system
TACAN	Tactical Air Navigation
TACSATCOM	Tactical Satellite Communications
TALLY HO	Visual contact
TANGO YANKEE	Thank You
TANK	Joint Chiefs of Staff conference room (also known as Gold Room)
TASKING	Selecting of reconnaissance targets
TAV	Trans-Atmospheric Vehicle
TECHINT	Technical Intelligence (collected by machines)
TELINT	Telemetry Intelligence
TELESCOPE	Submarine periscope displayed to signal start of exercise
TEXACO	Air refueling tanker
TIED ON	Aircraft is in formation
TIPS	External fuel tanks
TITS UP	Aircraft broken
TK	Talent-Keyhole (intelligence satellites)
TOE POPPERS	Land mines
TRAP	Arrested aircraft carrier landing
TRASH HAULER	Cargo aircraft or pilot
TRIDENT	US Navy ballistic missile
TRIPLE A	Anti aircraft fire
TRUE	Tactical reconnaissance in urban environment
TURN AND BURN	Sharp turn at maximum speed
TURKEY	F-14 Tomcat
TWEET	AT-37 attack jet
UHF	Ultrahigh Frequency, which is 200 MHz-3 GHz
USIB	United States Intelligence Board
VAMPIRE	Injured personnel on board
VECTOR	Turning aircraft
VERY HIGH	Altitude above 50,000 ft.
VERY LOW	Altitude below 500 ft.
VISITED	Target attacked
VFR	Visual Flight Regulations
VHF	Very High Frequency
VLF	Very Low Frequency
VMC	Visual Maintenance Condition
VULTURES ROW	Aircraft carrier observation deck
WARNING RED	Attack by hostile aircraft imminent
WARNING WHITE	Attack by hostile aircraft improbable
WARNING YELLOW	Attack by hostile aircraft probable
WARP ONE	Mach 1
WAVE OFF	Abort approach
WEAPONS FREE	Fire at any target not positively identified friendly
WEAPONS HOLD	Fire only in self-defense or to a formal order
WEAPONS LOOSE	Ready-fire at will
WEAPONS RED	(any tight) ready-do not fire for
CODEWORDS	

WEAPONS TIGHT
WEAPONS WHITE
WEAPONS YELLOW
WELL
WHALES
WHAT STATE
WHAT LUCK
WHAT'S UP
WHISKEY DELTA
WILCO
WILLY PETE
WINGMAN
WIZZO
WSO
YARDSTICK
ZIP LIP

ZIPPERS
ZOOMIE
ZULU TIME

political reasons
Fire only at hostile targets
(any tight) ready-fire in self-defense
(any tight) ready-fire with caution
Equipment operating efficiently
Navy refueling tankers
Fuel/ammunition/oxygen remaining
State results of mission
State problem
Lousy pilot (weak dick)
Will comply
White phosphorous (rockets)
Second pilot in two ship pair
Radar intercept operator
Weapons System Operator
Distance to target is
Routine transmission between air craft & controllers suspended
Combat air patrol target dawn and dusk
A jet jockey or aviator
Designation for Greenwich Meridian (mean)Time

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Intercepts Friends File

Review

The Yupiteru MVT-7100

Continuous-coverage handheld scanning receiver with SSB

By Howard Bornstein

Overview

The MVT-7100 is the latest offering in continuous-coverage handheld scanners by Yupiteru, coming on the heels of its very successful and well-received MVT-5000 and MVT-7000 scanning receivers. The 7100 adds substantial increase in capability over the 7000, primarily by dramatically increasing the number of scan channels, expanding the overall frequency coverage, and including sideband reception.

The MVT-7100 is a small, lightweight scanning receiver, offering continuous coverage from 100 KHz to 1.65 GHz! It will receive AM, FM, WFM (wide-FM), USB (upper sideband), LSB (lower sideband), and CW (continuous wave; i.e. Morse code) transmissions.

The 7100 has 1,000 scan channels organized in ten banks of 100 channels each and one additional dedicated channel for priority. It offers ten individual search banks with the ability to lock out up to 500 frequencies during a search. It can search in the following steps:

FM:

1 KHz, 5 KHz, 6.25 KHz, 9 KHz, 10 KHz, 12.5 KHz, 20 KHz, 25 KHz, 50 KHz, and 100 KHz.

Wide FM:

50 KHz, 100 KHz

AM:

1 KHz, 5 KHz, 6.25 KHz, 9 KHz, 10 KHz, 12.5 KHz, 20 KHz, 25 KHz, 50 KHz, and 100 KHz.

USB/LSB:

50 Hz, 100 Hz, 1 KHz, 5 KHz, 6.25 KHz, 9 KHz, 10 KHz, 12.5 KHz, 20 KHz, 25 KHz, 50 KHz, and 100 KHz.

The MVT-7100 is approximately 6.5 inches high, 2.5 inches wide and 1.5 inches deep, minus the antenna and belt clip. It is just slightly taller than the Radio Shack PRO-43, otherwise it is about the same size. The 7100 is a real lightweight, weighing in at only 11.5 ounces, without the antenna, but with the batteries and the belt clip.

Accessories

The MVT-7100 comes with an AC adaptor, a set of four nicad batteries (Hitachi 600 mAh), a car cigarette lighter adaptor, a belt clip, a wrist strap, an earphone, and a telescoping antenna with a pivot BNC mount.

Controls

The MVT-7100 is very cleanly laid-out. The top of the unit contains a BNC connector for the antenna, a knob for On/Off and volume, a knob for squelch control, and a tuning knob, which is used for a variety of functions.

On the left side of the unit are three recessed buttons: a momentary light button, a monitor button, and a keylock switch. There is also a small hole which contains a reset switch.

The Light button illuminates both the display and the keyboard with a very nice greenish light. Alas, it only stays on while you hold down the button. When will scanner manufacturers design their products with a method of letting the user keep the light on, especially when plugged into a power source?

The Monitor button immediately opens the squelch fully, allowing you to stop on a weak signal, without worrying about the scanner resuming its search or scan.

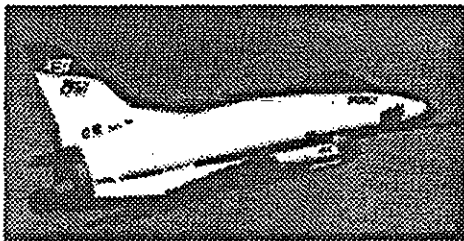
The Keylock performs the usual keyboard lockout of functions. It locks everything except the volume and squelch controls and the light and monitor buttons.

The MVT-7100 has a 15db attenuator feature built-in, but you select it with a keyboard combination, rather than with a separate button.

THE HIGH DESERT REPORT

By Ghost rider

Not only is the weather heating up here in the desert but so is the action in the R-2508 Complex. The NASA/Dryden Flight Research Facility has been filling the sky with its many exotic aircraft and one has even performed a revolutionary achievement. The X-31A Enhanced Fighter Maneuverability



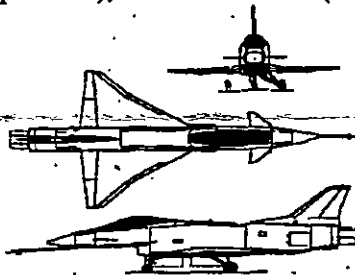
aircraft (EFM) (NASA 584, NASA 585) have set a major milestone by achieving a maneuver beyond the aerodynamic limits of any conventional aircraft. The new "Herbst Maneuver" (named after a German theorist) is a minimum radius, 180-degree turn using a post stall maneuver. In other words, it is able to turn on dime! The X-31 was designed for this radical maneuver, in which it can reverse course in 500 feet, one-fourth the distance required by any other military aircraft. For a better understanding, let's visit NASA/Dryden at Edwards AFB and take an X-31 for a ride! First a preflight check of the X-31 and the thrust vectoring paddles which will enable us to perform the maneuver. The X-31 is a single seat fighter, so while you strap into NASA-585, I will climb aboard today's chase F/A-18, NASA-847. Everything looks good, thumbs up, let's contact "CONFORM" on button-2 (304.000) for airspace coordination. The Edwards CP clears us to Edwards Ground, let's push button-3 (390.100). Winds are 270 at 06, altimeter 30.02, the sky is clear and cloud free, and we are cleared to taxiway Charlie and advised to contact Edwards Tower holding short of the active runway 22. Let's push button-4 (318.100). As we perform our final checks a KC-10 (LEAHI-09) roars past us and departs the runway for the Isabella AR track. The tower has cleared us for takeoff and we are on the roll. Gaining speed, a little back pressure on the stick and the X-31 seems to leap off the runway, well before the chase. Let's roll right heading 020 and push button-5 (272.000) to contact "SPORT CON-

TROL", since we will be working R-2515 today. "SPORT" clears us in and we advise we want to work a direct mission freq. of 283.700, let's switch as we are cleared to an altitude of 17,000 feet. Kick in the afterburner for a quick climb, whoa, wait for your chase! SPORT is calling for a radio check, go ahead and check in with them and NASA Mission Control (NASA 1). A few quick maneuvers to check the thrust-vectoring and it looks like we are ready! Cameras and tape on, accelerate to match 0.5 (200 Knots), full afterburner, you better suck that gut in, and prepare for a rapid pitch-up! Back stick pressure, feel those G's as the aircraft pitches up and you are almost perpendicular to the original flight path! The whole aircraft is now a huge speed brake as you rapidly decelerate. Suck some oxygen in and kick the paddles in to perform a rapid roll to the right as you reverse the direction of flight and accelerate back to high speed! That's it, you did it, now wait for me to catch up! Seems simple, but only the X-31 can do it at this time. The mission was a success, advise SPORT we are going to RTB, it's Miller time!

OTHER NASA NOTES

Dryden FRF has converted F/A-18B (NASA-845) from a chase bird to a new research aircraft designated SRA for Systems Research Aircraft.

It appears as if some funding for the three SR-71s has been cut back and I have not monitored any of the birds in the air recently, though interest is very high from many military and civilian organizations for one of a kind experiments. The birds on loan to NASA are SR-71B #64-17980 (NASA-831 two cockpit trainer), SR-71A #64-17971 (NASA-



832), and SR-71A #64-17980 (NASA-844).

Another unique aircraft has shown up on the NASA ramp lately, an ex-TACAMO C-130 is now sporting the NASA colors and a fuselage number of N427NA. Thanks to BIG RED for that observation.

a scan channel. This is due to the interaction of the tuning knob/arrow keys with the VFO. They let you change the current frequency up or down, based on the step value. An unfortunate side effect of the step value is that it rounds off a frequency so that it conforms to the current step value.

For example, if you enter 488.3375 MHz into the VFO, but the step value is 10 KHz, the frequency will automatically be converted to 488.330. In order to enter in 488.3375, you need to use a step of either 12.5 or 6.25 KHz. This isn't as bad as it may seem, however, because, once you do enter in the proper step, you can simply dial in the frequency you want. The display jumps by the step value when you the tuning knob or the arrow keys.

It turns out that for most frequency entry, this capability, along with the automatic advance to the next sequential scan bank, makes the MVT-7100 one of the easiest scanners around for entering frequencies—something you will appreciate when you try to fill up all 1,000 scan channels.

Scanning

Scanning on the MVT-7100 is a little unusual. If you just hit the SCAN button, the unit starts scanning the entire 1,000 channels sequentially. In order to scan a specific bank, you must first press the bank number and then the scan button. For example, to scan bank 5, you would press 5 SCAN. This is not so strange, except that

the unit doesn't remember which bank or banks you were scanning if you do anything to stop the scan. You have to press the bank number(s) each time or the scanner will try to scan the entire 1,000 channels, which is basically useless.

You may specify up to four banks to be scanned this way by entering in their numbers before pressing the SCAN button. For example, to scan banks 3,5, and 9, you would press 359 SCAN. While it seems strange to limit the number of banks to four, in practice this is not much of a problem, since you will rarely want to scan more than 400 channels at one time.

The arrangement of the memory as a 10 by 100 channel matrix is awkward because your smallest "chunk" of memory is 100 channels. However, the MVT-7100 has some additional features that allow you to better divide how you use these 1,000 channels.

First, you may lock out (it's called "Pass" in the MVT-7100) any channel in the scanner so that it isn't included in a bank scan. This works the way it does in most scanners, although there isn't a separate display indicator-for-lockout. Instead, the CH (channel) annunciator blinks on a channel that has been locked out.

The MVT-7100 has three different scan methods. The first is normal scan, which is as described above. You specify a bank and it scans all unlocked channels in that bank. You may also set up a programmed scan. You may specify up to ten channels per bank as "programmed" channels. When you do a programmed scan of a bank, only the programmed channels are scanned. This means that you can have a separate sub-group of ten channels per bank that are scanned independently. You perform a program scan like this:

nnnn FUNC P-SCAN (nnnn represents 1 to 4 bank numbers).

If you designate a channel as a programmed channel, it will be scanned in a program scan, even if the channel is locked out.

You may also specify a mode scan for up to four banks (or all ten banks if you don't specify any bank number). This will scan only the channels that are in the same mode as the VFO. You perform a mode scan like this: nnnn FUNC M-SCAN (nnnn represents 1 to 4 bank numbers).

These various scan options let you break the 100 channel bank into more useful sub-groups. For example, let's say that you put all 40 CB channels in bank 5 (401-440). CB is in AM mode. You then program your local PD into channels 450-470. They are in FM mode. Finally, you put in some railroad frequencies into channels 480-490. They are also in FM mode, but you mark them as "programmed" channels and also lock them out.

With this arrangement you can scan three independent groups, all in bank 5.

To scan the CB channel, set the VFO to AM mode and perform a mode scan:

5 FUNC M-SCAN

To scan the PD channels, set the VFO to FM mode and perform a mode scan: 5 FUNC P-SCAN (the railroad FM frequencies won't be included because they are locked out).

To scan the railroad frequencies, perform a programmed scan: 5 FUNC P-SCAN

While you probably wouldn't organize a bank like this, you get the picture.

One of the unusual aspects of the MVT-7100 is that you can't lock out scan banks. All the banks are always included in a scan unless you indicate the specific banks (up to 4) you want to scan. This is the way you will almost always perform a scan.

Searching

The MVT-7100 has ten separate search banks. Each search bank holds an upper and lower limit, a step increment, and a reception mode. You can change the step and the mode any time after setting up the search bank. You may also turn the attenuator on for the entire search bank.

Unlike scan banks, you may not link together several search banks.

A very useful feature in the MVT-7100 is the ability to lock out specific frequencies during a search. If you constantly stop on an open carrier, a data control channel, or just a noisy frequency, being able to lock out that frequency makes using the search features dramatically more useful, not to mention more enjoyable. You may also review the locked out frequencies and unlock them, if you wish.

The Keyboard

The keyboard is arranged in a 5 X 5 matrix. The keys are as follows:

1	2	3	SRCH
ATT	DELAY	SKIP	PRI
4	5	6	SCAN
SAVE	BEEP	PGM	MW
7	8	9	STEP
M-VFO	M-SCAN	P-SCAN	MODE

FUNC MODE (to change the mode).

Choose a mode with the arrows or turning knob.

STEP (to change the step value).

Choose a step value with the arrows or turning knob.

FUNC MW (to write the value back into the same channel).

Once you write a value to a scan channel, the scanner automatically steps to the next channel which makes sequential entry of frequencies extremely convenient.

It might seem unusual to have to indicate a step value when simply storing a frequency into CODEWORDS

Other Features

Skip

The MVT-7100 has a skip feature which is very much like the Seek feature found on some modern car radios. It can be used during either a search or a scan and will stop on an active frequency for 5 seconds and then continue on, even if there is still activity on the frequency.

Priority

The MVT-7100 dedicates a special channel (1000) as the priority channel. This channel is checked every 5 seconds when the priority feature is turned on and switches to it if there is activity on it.

Five seconds is really too long a wait for a priority check and you may miss short replies or the beginnings of communications due to this excessive wait. More typical priority delays are about 2 seconds.

Delay

The MVT-7100 has a delay feature but it is a bit confusing. The purpose of the delay is to wait a fixed time after a transmission ends before resuming the scan or search, because there may be a response and you will probably want to hear it. On the 7100, when then delay is *off*, the unit waits two seconds before resuming. When the delay is *on*, it waits four seconds! Again, this is probably due to the fact that Uniden holds the patents for variable timed delays on scanners and Yupiteru hasn't licensed them.

What this means is that you can never remove the delay entirely, which may make it a bit difficult to scan certain "trunked" systems. In these cases, the response may be on a different frequency than the original transmission, and if you wait around with the delay, you may miss it.

Save Mode

RFI sensitivity

The unit could be better shielded than it is. It definitely picks up radio frequency interference from computers and will lock up at a variety of different frequencies. You may need to be as far away as 20 feet from a computer to avoid this problem. I haven't noticed much RFI while operating the scanner in a car.

SSB operations

One of the special capabilities of the MVT-7100 is the ability to decode single sideband transmissions. Rather than requiring the use of a beat frequency oscillator (BFO), the 7100 uses true carrier injection to provide SSB reception. However, there are still a couple of things to consider. SSB signals are designed to be tuned 1.4 KHz above and below the carrier in the MVT-7100. So, instead of just dialing in a frequency, setting the mode USB, and getting the station, you need to tune above the desired frequency by some amount. While 1.4 KHz is the factory spec, each unit tends to be off by a certain amount. In addition, the accuracy may change as the unit warms up. However, once you "zero-in" your own receiver by watching how far off it appears, you will be able to accurately dial in the sideband frequencies you are interested in.

When you are sideband mode, you get two additional tuning increments: 50 Hz and 100 Hz. These very fine tuning steps make it extremely easy to tune in a station accurately.

As an extra benefit, some users have reported that by listening to speech inversion voice encryption in a sideband mode, they have been able to understand the transmissions!

Manual

The manual you get depends on where you obtain the scanner. Yupiteru provides an English translation of the Japanese manual, and it is, as expected, a bit hard to understand. Javiation has rewritten this manual in real English and it clearly describes all the operations of the MVT-7100. It is completely sufficient to get you started, although it doesn't go into detail about things like how organize you memory channels.

Modifications

There are no known modifications to this scanner, although it has only been available a short time. Unfortunately, when you open the unit, you automatically lose all of your programmed scan channels. This is somewhat of a distinctive to people experimenting with modifications or adding RFI shielding to the unit.

Recommendations

CODEWORDS

This is one of the most powerful, capable, and useful handheld wide-range scanning receivers ever built. The only other scanner with its features is the AOR AR1500 and the MVT-7100 stands heads and shoulders above it in terms of ease of use, sensitivity, and features. In one handheld unit, you have the capability of receiving virtually every type of voice transmission essentially across the entire radio spectrum! It is quite unbelievable. This unit is *highly recommended!*

Where to get it

The MVT-7100 has not (at the time of this review) received FCC type acceptance in the United States. It may or may not receive this acceptance before the deadline of April 26, 1993, which will preclude the importation of cellular-capable scanners. This means that there are currently no (and may never be) US importers of this scanner. However, a company in the UK, Javiation, has been able ship units to *individuals* in the US, as long as they are not for resale.

Jonathan Clough, owner of Javiation, has been available on CompuServe, particularly in HamNet, to take orders and support users. His support has been superb and he has been extremely helpful in answering questions, expedit-

ing orders, and keeping his customers apprised of changing circumstances. By the way, I have no personal interest in Javiation, except as a satisfied customer.

The price of the MVT-7100 will vary depending upon the relative strengths of the US dollar, the English pound, and the Japanese yen. However, it will probably fall in the range of \$550 or \$600 US. Price includes shipping by Federal Express. Jonathan is offering a one year warranty on units purchased through him.

He may be reached at:

Javiation
Carlton Works, Carlton Street
BRADFORD
West Yorkshire, BD7 1DA
United Kingdom

Phone 0274 732146 (From U.S. 011 44 274 732146)

Fax 0274 722627 (From U.S. 011 44 274 722627)

CompuServe address: 100117,535.

Please send any comments, corrections, or updates on this review to Howard Bornstein, CompuServe 76174,637.

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